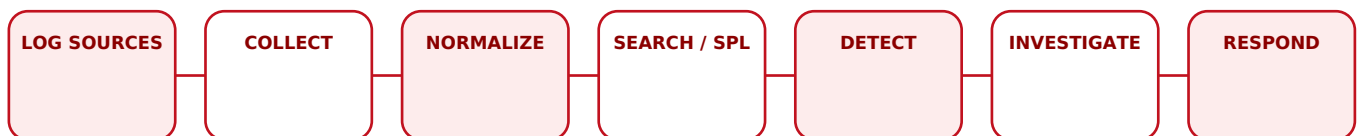


SPLUNK + SIEM

COMPLETE CYBERSECURITY MASTER COURSE

From absolute beginner → SOC Analyst → Detection Engineer → Advanced SIEM Practitioner

THE SIEM LIFECYCLE



PAW4NX • Learn • Grow • Build

Visual red + white study book • Labs • SPL • SOC workflows • Detection engineering • Interview preparation

Safety: practice only with systems, accounts, logs and networks you own or are explicitly authorized to test.

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01. SIEM Fundamentals

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What a SIEM is, what problems it solves, and how SOC teams use centralized telemetry.

CONCEPT MAP



Core knowledge

- Collect security telemetry
- Search and correlate events
- Detect suspicious behavior
- Support triage and investigation
- Provide evidence for response

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



02. SIEM Architecture

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The complete data path from source systems to analyst response.

CONCEPT MAP



Core knowledge

- Sources → collectors → parsing → indexing → search → detection → cases → response
- Plan for time sync, retention, availability, access control and auditability

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



03. Logs and Events

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Logs are records; events are observations; fields are the structured attributes analysts query.

CONCEPT MAP



Core knowledge

- Timestamp
- User/entity
- Host/device
- Source/destination
- Action/result
- Product and metadata

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

04. Security Data Sources

PAW4NX • SPLUNK + SIEM MASTER COURSE

Identity, endpoint, network, cloud, web, email and security-control telemetry.

CONCEPT MAP



Core knowledge

- Windows and Linux
- AD/Entra/SSO/MFA
- Firewall/DNS/proxy/VPN
- EDR
- Cloud audit logs
- WAF/web/application logs

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



05. Splunk Platform

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Splunk as the search, indexing, analytics and visualization foundation.

CONCEPT MAP



Core knowledge

- Splunk Enterprise
- Splunk Cloud Platform
- Splunk Enterprise Security
- Splunk SOAR

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



06. Splunk Vocabulary

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Indexes, sourcetypes, sources, hosts and knowledge objects.

CONCEPT MAP



Core knowledge

- Index = search/storage organization
- Sourcetype = event-data type/format
- Source = origin
- Host = generating system
- Lookup/macro/tag/event type = reusable context

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

07. Forwarders and Ingestion

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How telemetry moves into Splunk.

CONCEPT MAP



Core knowledge

- Collection design
- Transport
- Parsing
- Indexing
- Ingestion monitoring
- Failure handling

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

08. Parsing and Normalization

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Reliable timestamps and fields are prerequisites for reliable detections.

CONCEPT MAP



Core knowledge

- Event breaking
- Timestamp parsing
- Field extraction
- Normalization
- Data contracts
- Validation against raw events

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

09. SPL Foundations

PAW4NX • SPLUNK + SIEM MASTER COURSE

Learn SPL as a pipeline: filter → transform → aggregate → present.

CONCEPT MAP



Core knowledge

- search
- stats
- table
- eval
- where
- sort
- dedup
- rex
- lookup
- timechart

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=auth action=failure | stats count by user, src_ip | sort - count
```

10. Core SPL Commands

PAW4NX • SPLUNK + SIEM MASTER COURSE

The command set most analysts should know.

CONCEPT MAP



Core knowledge

- search/filter
- stats
- eval
- where
- table
- fields
- sort
- dedup
- rename
- rex
- lookup
- timechart

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=auth | stats count by user | sort - count | head 20
```

11. Time and Statistics

PAW4NX • SPLUNK + SIEM MASTER COURSE

Counts, distinct counts, rates, baselines and time windows.

CONCEPT MAP



Core knowledge

- Always validate time range
- Use time buckets
- Compare against baselines
- Separate volume from risk

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=auth | timechart span=5m count by action
```

12. Lookups and Macros

PAW4NX • SPLUNK + SIEM MASTER COURSE

Reusable enrichment and search logic.

CONCEPT MAP



Core knowledge

- Asset context
- Identity context
- Business exceptions
- Reusable search fragments
- Tags and event types

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

13. CIM and Data Models

PAW4NX • SPLUNK + SIEM MASTER COURSE

Normalize vendor telemetry into stable security concepts.

CONCEPT MAP



Core knowledge

- Common fields
- Data-model acceleration where appropriate
- Cross-source detections
- Field validation

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

14. Dashboards

PAW4NX • SPLUNK + SIEM MASTER COURSE

Operational views for SOC, management, hunting and data health.

CONCEPT MAP



Core knowledge

- Time range
- Filters
- KPIs
- Drilldowns
- Detection health
- Data health

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

15. Alerts

PAW4NX • SPLUNK + SIEM MASTER COURSE

Turn searches into actionable analyst work.

CONCEPT MAP



Core knowledge

- Clear title
- Entity
- Time
- Evidence
- Context
- Severity/risk
- Next action

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

16. Correlation and Detections

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Combine conditions and context into security findings.

CONCEPT MAP



Core knowledge

- Threat hypothesis
- Data prerequisites
- Logic
- Entity mapping
- Testing
- Tuning
- Ownership

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

17. Enterprise Security

PAW4NX • SPLUNK + SIEM MASTER COURSE

Security-focused detection, investigation and response workflows.

CONCEPT MAP



Core knowledge

- Detections
- Findings
- Investigations
- Response plans
- Threat intelligence
- Risk-based alerting

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



18. Mission Control

PAW4NX • SPLUNK + SIEM MASTER COURSE

Current Enterprise Security analyst workflow concepts.

CONCEPT MAP



Core knowledge

- Triage findings
- Group related activity
- Start investigations
- Add notes/evidence
- Follow response plans

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

19. Risk-Based Alerting

PAW4NX • SPLUNK + SIEM MASTER COURSE

Aggregate risk signals around entities rather than alerting on every low-level signal.

CONCEPT MAP



Core knowledge

- Risk events
- Risk objects
- Risk scores
- Risk factors
- Finding aggregation
- Thresholds

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



20. Entity Risk

PAW4NX • SPLUNK + SIEM MASTER COURSE

Users and assets as investigation anchors.

CONCEPT MAP



Core knowledge

- Identity context
- Asset priority
- Owner
- Role
- Known services
- Recent findings

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



21. Threat Intelligence

PAW4NX • SPLUNK + SIEM MASTER COURSE

Enrich observables such as IPs, domains, URLs and hashes.

CONCEPT MAP



Core knowledge

- Source
- Confidence
- Age
- Indicator type
- Context
- Corroboration

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



22. MITRE ATT&CK

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Use tactics and techniques to organize behavior and detection coverage.

CONCEPT MAP



Core knowledge

- Technique mapping
- Coverage gaps
- Detection intent
- Evidence vs mapping distinction

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



23. SOC Triage

PAW4NX • SPLUNK + SIEM MASTER COURSE

A repeatable process for deciding what needs deeper investigation.

CONCEPT MAP



Core knowledge

- Read alert
- Validate entity
- Build timeline
- Correlate
- Scope
- Escalate/contain
- Document

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



24. Authentication Analytics

PAW4NX • SPLUNK + SIEM MASTER COURSE

Detect abnormal login behavior.

CONCEPT MAP



Core knowledge

- Failures
- Success after failures
- MFA anomalies
- New devices
- Privileged activity
- Source anomalies

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=auth action=failure | stats count as failures by user, src_ip | where failures > 20
```

25. Brute Force and Password Spray

PAW4NX • SPLUNK + SIEM MASTER COURSE

Distinguish one-account guessing from many-account spraying.

CONCEPT MAP



Core knowledge

- Failures by user
- Users per source
- Time windows
- Service-account exceptions
- Post-success activity

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=auth action=failure | stats count as attempts, dc(user) as users by src_ip | sort - attempts
```

26. Suspicious Logins

PAW4NX • SPLUNK + SIEM MASTER COURSE

Use source, device, time, MFA and history together.

CONCEPT MAP



Core knowledge

- VPN/proxy context
- Device enrollment
- Travel context
- MFA result
- Follow-on activity

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=auth action=success | stats values(src_ip) values(device) by user
```

27. Privilege Abuse

PAW4NX • SPLUNK + SIEM MASTER COURSE

Monitor privileged group changes and administrative actions.

CONCEPT MAP



Core knowledge

- New admin accounts
- Group changes
- Role assignments
- Authentication changes
- Change-window correlation

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=auth | stats count values(action) by user, host
```

28. Endpoint Telemetry

PAW4NX • SPLUNK + SIEM MASTER COURSE

Processes, command lines, files, registry and network connections.

CONCEPT MAP



Core knowledge

- Process tree
- Parent/child
- Command line
- Persistence
- Network destination

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=endpoint | stats count values(parent_process) by process_name, user
```

29. PowerShell and Command Line

PAW4NX • SPLUNK + SIEM MASTER COURSE

Behavior-focused endpoint analytics.

CONCEPT MAP



Core knowledge

- Unusual parent process
- Encoded/obfuscated activity
- Destination context
- Automation allowlists
- Evidence preservation

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=endpoint process_name="powershell.exe" | stats count values(parent_process) by user, host
```

30. Network Analytics

PAW4NX • SPLUNK + SIEM MASTER COURSE

DNS, firewall, proxy, VPN and flow telemetry.

CONCEPT MAP



Core knowledge

- Destination rarity
- Beaconsing
- DNS anomalies
- Bytes
- Ports
- User context

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=network | stats sum(bytes) as bytes by src_ip, dest_ip | sort - bytes
```

31. Web Security

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Use web/WAF/application telemetry together.

CONCEPT MAP



Core knowledge

- HTTP method/status
- Client IP
- URI
- Auth endpoints
- Error spikes
- WAF correlation

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=web | stats count by uri, status | sort - count
```

32. Cloud Security

PAW4NX • SPLUNK + SIEM MASTER COURSE

Monitor identity, control-plane, workload and network activity.

CONCEPT MAP



Core knowledge

- Principal
- Action
- Resource
- Source
- Subscription/account/project
- Cloud findings

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=cloud | stats count by user, action, resource
```

33. Email Security

PAW4NX • SPLUNK + SIEM MASTER COURSE

Phishing investigation across email, endpoint and identity.

CONCEPT MAP



Core knowledge

- Sender/domain
- URL
- Click
- Endpoint process
- Mailbox activity
- Authentication

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=email | stats count values(url) by sender, recipient
```

34. Malware Investigation

PAW4NX • SPLUNK + SIEM MASTER COURSE

Pivot from file/process indicators into identity and network context.

CONCEPT MAP



Core knowledge

- Hash/path
- Process tree
- Persistence
- DNS
- C2
- Affected hosts

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=endpoint | stats values(process_name) values(dest_domain) by file_hash
```

35. Data Exfiltration

PAW4NX • SPLUNK + SIEM MASTER COURSE

Analyze volume, destination and behavior together.

CONCEPT MAP



Core knowledge

- Large transfers
- Rare destinations
- Off-hours
- Sensitive resources
- Business exceptions

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=network | stats sum(bytes) as outbound by user, dest_domain | sort - outbound
```

36. Threat Hunting

PAW4NX • SPLUNK + SIEM MASTER COURSE

Hypothesis-driven proactive investigation.

CONCEPT MAP



Core knowledge

- Hypothesis
- Data
- Pivot
- Search
- Correlate
- Validate
- Document

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=* | stats count by user, host, action | sort - count
```

37. Detection Engineering

PAW4NX • SPLUNK + SIEM MASTER COURSE

Build detections as maintainable analytical products.

CONCEPT MAP



Core knowledge

- Threat model
- Data contract
- SPL
- Tests
- Deployment
- Monitoring
- Tuning

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=auth | stats count by user, src_ip | sort - count
```

VISUAL WORKFLOW



38. Detection Quality

PAW4NX • SPLUNK + SIEM MASTER COURSE

Reduce noise without suppressing meaningful threats.

CONCEPT MAP



Core knowledge

- False positives
- Exceptions
- Thresholds
- Time windows
- Deduplication
- Metrics

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=* | stats count by detection_name | sort - count
```

VISUAL WORKFLOW



39. Incident Response

PAW4NX • SPLUNK + SIEM MASTER COURSE

Use SIEM to detect, scope, timeline and document incidents.

CONCEPT MAP



Core knowledge

- Detect
- Triage
- Scope
- Contain
- Eradicate
- Recover
- Lessons learned

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



40. SOAR

PAW4NX • SPLUNK + SIEM MASTER COURSE

Automate repetitive enrichment and controlled response.

CONCEPT MAP



Core knowledge

- Enrichment
- Decision
- Playbook
- Approval
- Action
- Audit

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



41. Performance and Cost

PAW4NX • SPLUNK + SIEM MASTER COURSE

Search efficiently and manage data lifecycle.

CONCEPT MAP



Core knowledge

- Narrow time ranges
- Target indexes
- Filter early
- Use acceleration appropriately
- Retention by value
- Monitor ingestion

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



42. SIEM Access Control

PAW4NX • SPLUNK + SIEM MASTER COURSE

Protect security telemetry and administrative capabilities.

CONCEPT MAP



Core knowledge

- RBAC
- MFA
- Audit
- Data masking
- Change control
- Secrets management

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

43. Troubleshooting

PAW4NX • SPLUNK + SIEM MASTER COURSE

A no-results search is usually a data/time/field problem before it is an SPL problem.

CONCEPT MAP



Core knowledge

- Check time
- Check index
- Check sourcetype
- Inspect raw events
- Check fields
- Check ingestion

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=auth action=failure | stats count by src_ip | sort - count
```

44. Common SPL Patterns

PAW4NX • SPLUNK + SIEM MASTER COURSE

Reusable analyst query patterns.

CONCEPT MAP



Core knowledge

- Top failures
- Activity by user/host
- Time trends
- Distinct counts
- Correlations

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=* earliest=-24h | stats count by sourcetype
```

45. Hands-on Lab

PAW4NX • SPLUNK + SIEM MASTER COURSE

Build a safe home SOC.

CONCEPT MAP



Core knowledge

- Isolated network
- SIEM
- Windows VM
- Linux VM
- Synthetic/test activity
- Dashboards and detections

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

Practical SPL / investigation pattern

These are learning templates. Replace field names with fields actually present in your authorized lab data.

```
index=auth | timechart span=15m count by action
```

46. Lab Progression

PAW4NX • SPLUNK + SIEM MASTER COURSE

Move from searches to advanced detection engineering.

CONCEPT MAP



Core knowledge

- Raw search
- Statistics
- Dashboards
- Alerts
- Lookups
- CIM
- Detections
- RBA
- Hunting
- Automation

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



47. Job-Ready SOC Workflow

PAW4NX • SPLUNK + SIEM MASTER COURSE

Practice the full analyst loop.

CONCEPT MAP



Core knowledge

- Alert
- Validate
- Pivot
- Scope
- Classify
- Respond
- Document

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



48. Interview Preparation

PAW4NX • SPLUNK + SIEM MASTER COURSE

Model questions and answers for SOC/SIEM roles.

CONCEPT MAP



Core knowledge

- SIEM
- Splunk
- SPL
- CIM
- RBA
- Detection
- Threat hunting
- Tuning

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



49. 30/60/90 Roadmap

PAW4NX • SPLUNK + SIEM MASTER COURSE

A structured plan for building job-ready capability.

CONCEPT MAP



Core knowledge

- Days 1-30: fundamentals
- 31-60: detections/investigation
- 61-90: hunting/detection engineering

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



50. Quick Revision

PAW4NX • SPLUNK + SIEM MASTER COURSE

A compact mental model.

CONCEPT MAP



Core knowledge

- Data → fields → SPL → context → detection → finding → case → response

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



51. Glossary

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Core SIEM and SOC vocabulary.

CONCEPT MAP



Core knowledge

- SIEM
- SPL
- CIM
- RBA
- IOC
- TTP
- EDR
- NDR
- SOAR
- UEBA
- MFA

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

52. Portfolio Projects

PAW4NX • SPLUNK + SIEM MASTER COURSE

Demonstrate reasoning, not just screenshots.

CONCEPT MAP



Core knowledge

- Authentication SOC
- Endpoint SOC
- Network SOC
- Cloud SOC
- Threat hunting
- Detection engineering
- RBA

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

53. Advanced Practitioner

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What comes after basic SPL.

CONCEPT MAP



Core knowledge

- Data quality
- Normalization
- Analytics
- Risk
- Detection lifecycle
- Case management
- Automation

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



54. Current Enterprise Security

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Version-aware concepts for modern Enterprise Security.

CONCEPT MAP



Core knowledge

- Findings/finding groups
- Detection editor
- Risk-based alerting
- Mission Control
- Response plans

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



55. Official Learning

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How to keep your knowledge version-correct.

CONCEPT MAP



Core knowledge

- Use official documentation
- Match your exact version
- Validate UI/commands
- Review release notes

Analyst mindset

Always verify the data source, time range, field quality and business context before treating a search result as evidence. A detection is a hypothesis supported by telemetry; the investigation determines what the evidence means.

VISUAL WORKFLOW



SPL DEEP DIVE: STATS

ADVANCED STUDY CARD

stats is the core aggregation command for many SOC searches.

VISUAL MODEL



- count gives event volume; dc(field) gives distinct values; values(field) collects values; sum/avg/min/max support quantitative analysis.
- Example: index=auth | stats count as failures, dc(user) as users by src_ip | sort - failures

Practice rule

Repeat the concept in an authorized lab until you can explain the evidence, assumptions and limitations without relying on memorized screenshots.

SPL DEEP DIVE: EVAL + WHERE

ADVANCED STUDY CARD

Create calculated fields, then filter on them.

VISUAL MODEL



- Example: `index=network | eval mb=round(bytes/1024/1024,2) | where mb>100 | table _time user dest_ip mb`
- Validate units and assumptions before using a threshold.

Practice rule

Repeat the concept in an authorized lab until you can explain the evidence, assumptions and limitations without relying on memorized screenshots.

SPL DEEP DIVE: REX

ADVANCED STUDY CARD

Use rex for controlled extraction when the source does not already provide the needed field.

VISUAL MODEL



- Prefer structured fields when available. Regex can be brittle if the raw format changes.
- Example pattern: `| rex field=_raw "user=(?[^ ]+)"`

Practice rule

Repeat the concept in an authorized lab until you can explain the evidence, assumptions and limitations without relying on memorized screenshots.

SPL DEEP DIVE: LOOKUP

ADVANCED STUDY CARD

Enrichment converts technical identifiers into operational context.

VISUAL MODEL



- Example: | lookup asset_lookup ip as src_ip OUTPUT owner priority category
- Validate lookup freshness, duplicate keys and ownership.

Practice rule

Repeat the concept in an authorized lab until you can explain the evidence, assumptions and limitations without relying on memorized screenshots.

DETECTION SPEC TEMPLATE

ADVANCED STUDY CARD

A production detection should be documented.

VISUAL MODEL



- Name • hypothesis • ATT&CK; mapping • data sources • required fields • search logic • schedule • threshold • severity/risk • false positives • exceptions • response • owner • test cases • version.

Practice rule

Repeat the concept in an authorized lab until you can explain the evidence, assumptions and limitations without relying on memorized screenshots.

INVESTIGATION TIMELINE

ADVANCED STUDY CARD

Build a timeline before deciding what happened.

VISUAL MODEL



- Anchor on the first suspicious event. Pivot backward for initial access and forward for persistence, privilege, lateral movement, collection and exfiltration signals.
- Use UTC/local-time awareness and preserve original timestamps.

Practice rule

Repeat the concept in an authorized lab until you can explain the evidence, assumptions and limitations without relying on memorized screenshots.

FALSE POSITIVE ENGINEERING

ADVANCED STUDY CARD

Tune with evidence, not guesswork.

VISUAL MODEL



- Identify why the alert is firing. Decide whether behavior is expected, whether the exception is stable, and whether the exception can be scoped narrowly.
- Use expiration and ownership for temporary exceptions.

Practice rule

Repeat the concept in an authorized lab until you can explain the evidence, assumptions and limitations without relying on memorized screenshots.

DATA QUALITY SCORECARD

ADVANCED STUDY CARD

A detection is only as good as its inputs.

VISUAL MODEL



- Check ingestion continuity, timestamp correctness, field completeness, normalization, duplication and source ownership.
- Track broken sources as operational findings.

Practice rule

Repeat the concept in an authorized lab until you can explain the evidence, assumptions and limitations without relying on memorized screenshots.

SOC CASE NOTE TEMPLATE

ADVANCED STUDY CARD

Write investigations so another analyst can reproduce your reasoning.

VISUAL MODEL



- Summary • affected entity • first/last observed • evidence • pivots • scope • benign explanation considered • actions taken • remaining risk • escalation • closure reason.

Practice rule

Repeat the concept in an authorized lab until you can explain the evidence, assumptions and limitations without relying on memorized screenshots.

INTERVIEW RAPID FIRE

ADVANCED STUDY CARD

Practice concise answers.

VISUAL MODEL



- What is SIEM? Centralized security telemetry plus search/analytics and detection/investigation workflows.
- What is SPL? Splunk's search and transformation language.
- What is RBA? Aggregation of risk signals around entities to support higher-confidence findings.
- What is CIM? A common vocabulary/data model for consistent security analytics.

Practice rule

Repeat the concept in an authorized lab until you can explain the evidence, assumptions and limitations without relying on memorized screenshots.

FINAL MASTER REVISION

PAW4NX • Learn • Grow • Build

THE COMPLETE SIEM LOOP

MASTER MENTAL MODEL



Job-ready goal: take a security finding, validate the data, build a timeline, correlate identity/endpoint/network context, explain why it matters, document evidence, and recommend the next authorized response step.

Safety: security tools and techniques must be used only on systems and data you own or are explicitly authorized to test.

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